

2024 Ph H2 Q10

Section: Particles and Waves

Topic: Interference of Sound

Summary of Question

Two loudspeakers are driven in phase by the same signal generator (frequency 1700 Hz). A microphone moves from X to Y and detects maxima/minima. At point P the distances are: 1.80 m from loudspeaker 1 and 1.50 m from loudspeaker 2.

Tasks: (a) Calculate the wavelength. (b) Determine whether interference at P is constructive or destructive. (c) State and justify the effect on the amplitude at P if loudspeaker 1 is disconnected.

(a) Wavelength of the sound

Relationship: $\lambda = v / f$.

Substitution: $\lambda = 340 / 1700 = 0.200 \text{ m}$.

Final answer: 0.200 m

(b) Type of interference at point P

Path difference: $\Delta = |r_1 - r_2|$.

$$\Delta = |1.80 - 1.50| = 0.30 \text{ m.}$$

Compare with $\lambda = 0.200 \text{ m}$: $\Delta / \lambda = 1.5 = 1.5 \rightarrow \Delta = (1 + \frac{1}{2})\lambda$.

Condition $\Delta = (m + \frac{1}{2})\lambda$ gives a minimum (destructive interference).

Final answer: destructive interference at P

(c) Disconnect loudspeaker 1 — effect at P

With one loudspeaker disconnected, there is no interference pattern; only a single source remains.

At P the amplitude rises from (near) zero (destructive interference) to the amplitude produced by the single operating loudspeaker at its distance.

Final answer: amplitude increases (no cancellation with a second source).

Revision Tips

Two coherent sources in phase: maxima when $\Delta = m\lambda$;
minima when $\Delta = (m + \frac{1}{2})\lambda$.

Always compute Δ/λ to classify the point quickly.

Turning off one source removes interference: the microphone detects a single-source signal only.